

Replayable BRAVO Ballot-Polling Audit Transcripts

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Abstract

BRAVO is the classic ballot-polling risk-limiting audit: it samples paper ballots, records whether each observation supports the reported winner or loser, and updates a sequential likelihood ratio until the reported outcome has enough evidence to stop or the audit escalates. This paper documents the RCOUNT BRAVO contract, which separates the statistical claim from its public evidence. A verifier reads the reported two-candidate totals and risk limit from an audit record, reconstructs each ballot observation from declared sample steps, replays the BRAVO likelihood ratio in integer parts-per-million arithmetic, and confirms that the declared per-draw statistic and stop decision follow from the public inputs. The contract is the simplest statistically meaningful bridge from sampled ballot observations to a replayable risk-limit transcript, and it is exposed through the same algorithm-replay surface as the other RCOUNT audit methods.

1 Introduction

Ballot-polling risk-limiting audits are usually presented as a statistical test: sample ballots at random, update a likelihood ratio, and stop when the reported winner is supported at the chosen risk limit. As with sampler replay (V.7), a large part of public trust sits one layer below that test. An observer should be able to ask whether the declared per-draw statistic and stop decision actually follow from the reported totals, the risk limit, and the sequence of observed ballots.

RCOUNT therefore treats BRAVO as a replay contract rather than a number to be trusted. The verifier does not re-run a vendor audit tool or accept an exported risk value. It reconstructs each observation from the public audit record and recomputes the BRAVO likelihood ratio in integer arithmetic, so the published transcript can be regenerated from public evidence.

This paper covers the BRAVO ballot-polling slice. It does not claim that any particular election was audited correctly, nor that BRAVO is the right method for a given jurisdiction. It claims that, when an artifact contains ballot-polling observations and reported two-candidate totals, the resulting stop-or-continue transcript is reproducible.

2 BRAVO Contract

The audit record carries the public inputs the BRAVO replay needs:

Field	Meaning
<code>method_id</code>	Versioned method name. Current value: <code>bravo-ballot-polling-v1</code> .
<code>risk_limit_ppm</code>	Risk limit in parts per million, avoiding floating-point drift.
<code>reported_winner_votes</code>	Reported votes for the winner of the two-candidate assertion.
<code>reported_loser_votes</code>	Reported votes for the loser of the assertion.
<code>sample_steps</code>	Ordered ballot observations, each a rational mark used to classify the draw.

Each sample step is reduced to one of three BRAVO observations for the audited assertion. A mark of 1/1 is a vote for the winner, 0/1 is a vote for the loser, and any other value with a positive denominator is a non-vote or other-candidate ballot that carries no evidence for the two-candidate test.

The replay walks the observations in order. From the reported totals it forms the winner and loser shares of the two-candidate population and updates the sequential likelihood ratio after each draw: a winner observation multiplies the statistic by twice the reported winner share, a loser observation multiplies it by twice the reported loser share, and an other observation leaves it unchanged. The audit stops the first time the likelihood ratio reaches the risk-limit threshold $1/\alpha$, where α is the declared risk limit. All comparisons are carried in integer parts-per-million so that two independent verifiers reach the same per-draw statistic and the same stop index.

3 Implemented Surface

The BRAVO replay is dispatched by method id alongside the other RCOUNT audit algorithms. When an audit record names `bravo-ballot-polling-v1`, the algorithm-replay surface routes it to `replay_bravo_run` in `rcount-audit`. That function enforces the contract before any statistic is computed: a missing risk limit, missing reported winner votes, or missing reported loser votes each produces a boundary transcript that preserves the declared claim without replaying it. The same boundary outcome is produced when a sample step cannot be reduced to a valid BRAVO observation.

Once the inputs are present, `replay_bravo_run` maps every declared sample step to a winner, loser, or other observation and calls `replay_bravo_ballot_polling` in `rcount-stats`. The statistics crate owns the sequential likelihood-ratio recurrence and returns, for each draw, the running statistic, a risk estimate in parts per million, and whether the audit has stopped. The audit crate then zips those computed steps back against the declared steps to build the replay transcript.

Check	Failure caught
<code>audit_algorithm_run</code>	Unsupported method id, or BRAVO record missing risk limit, reported winner votes, or reported loser votes.
<code>bravo_replay</code>	Declared per-draw likelihood ratio, risk estimate, or stop decision drifts from the recomputed transcript.
<code>cvr_summary_total</code>	Reported contest totals still reconcile to the public summaries that anchor the assertion.
<code>source_hash_match</code>	Source evidence bytes still match the declared source index.

Because BRAVO shares the algorithm-replay dispatch, its failures surface through the same command and transcript shape as sampler replay, stopping checks, and the other ballot-polling and comparison methods. An observer sees the declared and recomputed statistics side by side rather than a single trusted verdict.

4 Fixtures

The positive fixture is a tiny two-candidate run. It reports the winner ahead three to one, sets the risk limit to ten percent, and supplies a short sequence of winner observations. The likelihood ratio climbs past the risk-limit threshold and the replay reports a stop; the declared per-draw statistics match the recomputed transcript, so the package passes. This is the toy run exercised by the `bravo_algorithm_statistics_replay_passes_toy_run` test.

The continue fixture uses the same reported totals but a sequence of observations that does not accumulate enough evidence. The likelihood ratio never reaches the threshold, so the transcript reports continue rather than stop, and the declared statistic must agree with that non-stopping path. This is the case covered by `bravo_algorithm_statistics_replay_continues_when_th`

The drift negative fixture keeps the observations and totals fixed but edits a declared per-draw statistic or stop decision. The recomputed transcript diverges from the declared one, and the package fails: the transcript exposes the declared and computed values for the offending draw.

The missing-field boundary fixture omits a required reported vote total or the risk limit. The audit record is preserved, but the BRAVO claim is not replayed: the verifier emits a boundary transcript that explicitly records the public evidence as insufficient to recompute the statistic, rather than silently passing or failing.

Each fixture pair separates the two failure modes that matter for a ballot-polling audit: an audit that does not actually stop under its declared observations, and an audit that cannot be replayed at all from the public record.

5 Boundaries

This slice verifies a single ballot-polling outcome assertion under sampling assumptions. BRAVO confirms that a declared stop-or-continue transcript follows from reported two-candidate totals, a risk limit, and an ordered list of ballot observations. It does not verify ballot custody, voter eligibility, or any ballot-level CVR comparison claim; those live in other RCOUNT contracts.

The statistical boundary is also deliberate. The implemented recurrence is the classic two-candidate BRAVO likelihood ratio in integer parts-per-million. It is not Minerva, ALPHA, Kaplan-Markov, or a stratified or comparison risk measure, and it does not decide whether a jurisdiction’s legal RLA method permits replacement sampling, round-by-round planning, or multi-candidate assertions. Multi-candidate contests must be reduced to the relevant pairwise assertions before this contract applies.

Finally, BRAVO replay assumes the sample order is already accounted for. It checks that the observed marks support the declared decision, not that the ballots were the ones a public seed should have selected; that reproducibility claim is the separate sampler-replay contract of V.7. As with every RCOUNT audit slice, the public-evidence privacy boundary remains with the jurisdiction: publishing per-ballot observations can expose rare ballot styles, so small-batch publication rules are an external responsibility.

6 Conclusion

RCOUNT now has a replayable BRAVO ballot-polling contract. The verifier reads the reported two-candidate totals and risk limit, reconstructs each ballot observation from the declared sample steps, and recomputes the sequential likelihood ratio in integer parts-per-million, so the published stop-or-continue transcript can be regenerated from public evidence. Required-field gaps produce an explicit boundary transcript rather than a trusted verdict, and the method is exposed through the same algorithm-replay surface as the other RCOUNT audit checks.

BRAVO is the simplest statistically meaningful step from sampled observations to a risk-limit transcript, and it establishes the pattern that the round-based (Minerva/Athena), betting-martingale (ALPHA), and comparison (Kaplan-Markov, batch, SHANGRLA) methods extend. With ballot-polling replay in place, later work can focus on the distinct statistical machinery of those methods without again re-establishing how a declared transcript is tied back to public inputs.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Replayable risk-limiting audit samplers and stopping checks. Technical report, Apportionment Research, 2026b.